Weekly Update, 2–6 Nov 2026

Brian Zhou · CSE 5999 hiking-speed project · for Danni Beaulieu · 6 Nov 2026

This week all five Gate 2 checks passed, I hand-checked the stop rule, and I tested aerial imagery. Neither ready-made imagery embeddings (Track A) nor photos centred on the trail (Track B) carried over to new regions; terrain still does.

This week at a glance

Mon 2 Nov	Wed 4 Nov	Thu 5 Nov	Fri 6 Nov
Architecture review learned Tobler's setup, explained to Danni	**Gate 2 brief** five checks and the imagery plan	**Stop-rule check** 20 GPS moments labeled by hand	**Track A tested** helps in known places, not in new regions
		Clay imagery 754 tiles checked, 6 GB downloaded	**Track B tested** 164,415 trail photos: same result as A

Gate 2: all five checks pass

Gate 2 check	Result
Every learned model beats Tobler on held-out hikes	Yes: Tobler's fixed numbers are the worst model in every test
Trail tags beat slope alone	Yes: +1.3–1.5% on new hikes
Remote sensing compared with tags, gap measured	In unseen regions terrain adds 2.1–2.8%, tags 0.4–0.9%; terrain wins in 16 of 17 regions
Uncertainty is calibrated	The 90% ranges hold 89.4–89.9% of test segments
Our model beats boosted trees on the same inputs	A tie with the same spread rule; +0.55–0.66 points with its own spread

In new regions, learned Tobler is 3.4% better than slope alone; Tobler is 12.5% worse

CRPS skill vs a slope-only curve, each of 17 regions held out in turn; best version of each model



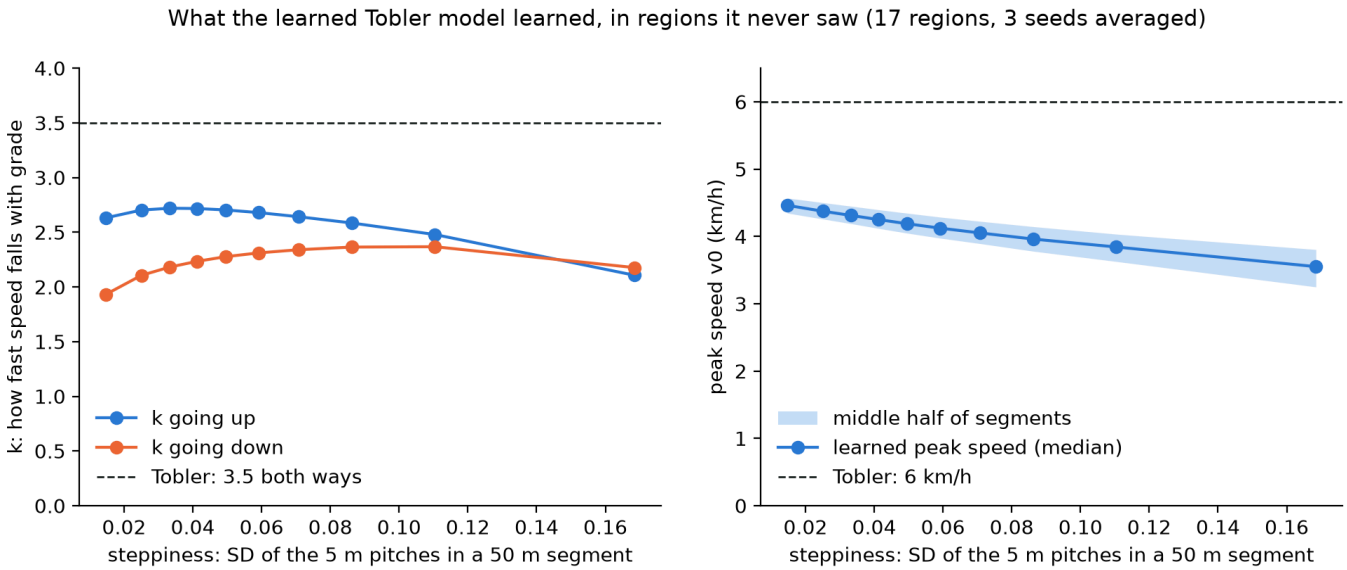
Source: gate2_summary.csv and the paper's Table I. Tags: additive model; terrain: boosted trees; learned Tobler: its own spread.

The one caveat is check 5: learned Tobler's edge comes from its own uncertainty, which the trees did not have in this test.

What learned Tobler learned

Its constants change with the terrain in sensible ways. These are read in regions it never saw (17 regions, 3 seeds averaged):

- **k**, how fast speed falls as the trail steepens, stays between about 1.9 and 2.7, never Tobler's 3.5.
- **Peak speed** falls from 4.5 km/h on smooth trail to 3.6 km/h on steppy trail (Tobler: 6).
- Going down smooth trail is easy (k down 1.9); steppy trail slows descents too.



Left: k going up and down; right: peak speed. Dashed: Tobler's 3.5 and 6 km/h.

Stop-rule hand check

The rule that built our dataset misses most short stops; a plain speed rule matches my hand labels better. I labeled 20 GPS moments from White Mountains hikes as stopped or moving (19 usable answers: 7 stops, 12 moving).

Stop rule	Agrees with me	Stops caught	False stops
Dataset rule: 60 s ending < 10 m away	8 of 19	1 of 7	5
Speed rule: < 0.5 km/h	13 of 19	6 of 7	5
Either rule	12 of 19	6 of 7	6

The dataset rule cannot see stops under a minute, and most of mine were 20–40 s. Switching to the speed rule barely moves the White Mountains curve: peak speed 4.23 → 4.13 km/h, k 2.10 → 2.15. My plan: keep the paper's numbers, state this check as a limitation, and use the speed rule in the data release.

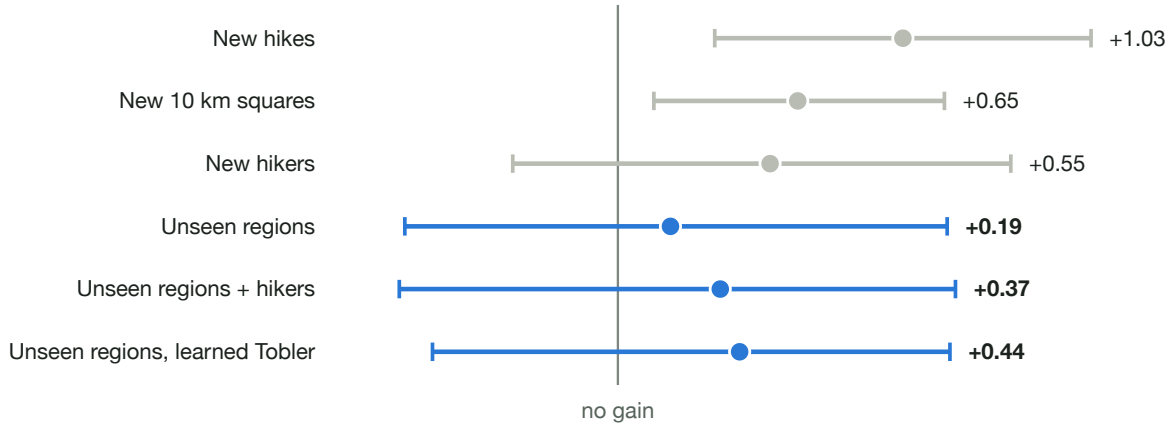
Imagery Track A: helps where the model has been, not in new regions

Track A used Clay v1.5's ready-made embeddings: 1,024 numbers for every 155 m square of NAIP aerial photos, already computed. Each walking segment took the square under its midpoint. That covers 15 regions:

Clay has Olympic only as 2017 photos at 1 m, and the North Shore is in Canada. In each test fold, PCA shrank the numbers to 16, fitted on that fold's training rows only, and they were added to the Gate 2 models.

Imagery helps where the model has been (+1.0 on new hikes), not in new regions (+0.2)

Gain from adding Clay imagery to terrain, in CRPS skill points, with 95% range; 15 regions



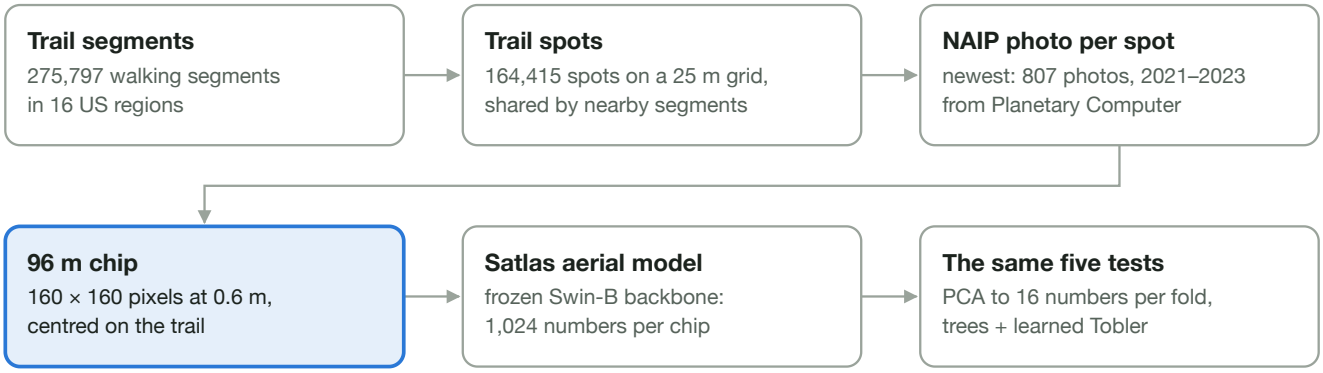
Source: results/t3_imagery/track_a_trees_summary.csv and the learned-Tobler check (decision #125). Trees unless noted.

By the rule I set before running it (a gain above zero in the new-region tests), Track A does not pass. It is the same pattern as the satellite embeddings: the model uses imagery to recognize places it has seen, not to learn trail features that carry over.

Imagery Track B: photos centred on the trail, same result

Track A's 155 m squares mostly show the landscape around a trail. Track B tests whether a closer look at the trail itself carries over to new regions, with the same tests and the same pass rule.

Track B looks at the trail itself: a 96 m photo centred on every trail spot



Decision #126; scripts t3_naip_spots.py and t3_naip_satlas.py.

Result (6 Nov): Track B does not pass either. Photos of the trail itself also help only where the model has been. Skill in % over the slope-only curve, 16 regions, trees unless noted:

Test	Terrain	Terrain + imagery	Gain [95% range]
New hikes	3.20	4.21	+1.01 [+0.57, +1.44]
New 10 km squares	2.79	3.24	+0.45 [+0.01, +0.85]
New hikers	3.10	3.62	+0.52 [−0.08, +1.09]
Unseen regions	2.92	3.08	**+0.16 [−0.24, +0.57]**
Unseen regions + hikers	2.52	2.43	**−0.09 [−0.79, +0.46]**
Unseen regions, learned Tobler	3.50	3.50	**−0.01 [−0.52, +0.34]**

So neither landscape-scale nor trail-centred aerial imagery carries over to new regions, while terrain does.

What the model sees. Two of the 20 photos I checked by hand, with our matched trail in red: under forest the trail is hidden; in open terrain the path is visible.



White Mountains, forest: the trail (red) is hidden under the canopy.



Front Range, alpine meadow: a faint path runs along the red line.

Next steps

- **Tue 10 Nov:** the IGARSS submission site opens; the paper moves to the official template.
- **Wed 11 Nov:** decide how imagery goes in the paper; both tracks point to one sentence: it helps within regions, not in new ones.
- **16–22 Nov:** the full draft, rewritten in my own words.